Points to Remember

- Rutherford's alpha-particle scattering experiment led to the discovery of the atomic nucleus.
- J.Chadwick discovered the presence of neutrons in the nucleus.
- Mass number of an element is the total number of protons and neutrons.
- Valence electrons are the electrons in the outermost orbit.
- Valency is the combining capacity of an atom.
- Isotopes are atoms of the same element, which have same atomic number but different mass numbers.
- Isobars are the atoms of the different element with same mass number but different atomic number.
- Isotones are the atoms of different elements having same number of neutron but different atomic number and mass number.

GLOSSARY

Atom	The smallest component of an element which takes part in a chemical reaction.
Electron	A stable subatomic particle with a charge of negative electricity, found in all atoms and acting as the primary carrier of electricity in solids.
Neutron	A subatomic particle of about the same mass as a proton but without an electric charge, present in all atomic nuclei except those of ordinary hydrogen.
Orbitals	Atomic orbitals are region of space around the nucleus of an atom where an electron is likely to be found.
Proton	A stable subatomic particle occurring in all atomic nuclei, with a positive electric charge equal in magnitude to that of an electron.
Quantum numbers	The numbers which designate and distinguish various atomic orbitals and electrons present in an atom.

TEXTBOOK EXERCISES

I. Choose the correct answer.

1. Among the following the odd pair is
   a) $^{18}_{8}O$, $^{19}_{9}F$ b) $^{40}_{18}Ar$, $^{14}_{7}N$
   c) $^{30}_{14}Si$, $^{31}_{15}P$ d) $^{40}_{20}Cr$, $^{39}_{19}K$
2. Change in the number of neutrons in an atom changes it to
   a) an ion. b) an isotope.
   c) an isobar. d) another element.
3. The term nucleons refer to
   a) protons and electrons
   b) only neutrons
   c) electrons and neutrons
   d) protons and neutrons
4. The number of protons, neutrons and electrons present respectively in $^{80}_{35}Br$ are
   a) 80, 80, 35
   b) 35, 55,80
   c) 35, 35, 80
   d) 35, 45, 35
5. The correct electronic configuration of potassium is
   a) 2,8,9 b) 2,8,1
   c) 2,8,8,1 d) 2,8,8,3

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